

Statement of Work (SOW) - Data Lake Implementation Project

Data Lake House Implementation on AWS for Advanced Analytics

Project developed for

Caleb S.A. de C.V.

Implemented by

Team 6 - Data Engineering, UNAM 2026.

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1. Project Context

Caleb S.A. de C.V. requires a data engineering solution to process and analyze the historical match records of national football teams from the external API-Football, specifically those directly associated with the Mexican national football team. The lack of a centralized architecture hinders the generation of reliable statistical KPIs on the international performance of national teams, making it difficult to identify areas of opportunity for the Mexican team.

2. Project Summary

The objective of this project is to design and build a Data Lake architecture based on Amazon Web Services (AWS) that enables scalable storage in Amazon S3 under a three-layer architecture (Bronze, Silver, and Gold), with analytical processing through AWS Glue and Amazon Athena.

The solution consolidates data in Parquet format, partitioned by team and year, dependent on the AWS Glue catalog and queried via standard SQL through Amazon Athena, enabling data-driven decision-making based on accurate information about the sporting performance of the analyzed teams.

3. Scope of Work (SOW)

The consulting team will be responsible for the following activities:

3.1 Data Ingestion

Configuration of the data extraction pipeline from API-Football to the *Landing* zone of Amazon S3. Files will be stored in .csv format at the following path:

s3://unam-2026-ingenieriadatos-equipos6/1bronze/

3.2 Storage and Organization

Implementation of a three-layer architecture in the bucket:

unam-2026-ingenieriadatos-equipos6

- **Bronze (/1bronze/):**
Raw data directly extracted from the API, without transformation.
 - **Silver (/2silver/):**
Clean data, partitioned in Parquet format. Path: /2silver/[team]/year=YYYY/
 - **Gold (/3gold/):**
Aggregated views with high-value business KPIs ready for analytical consumption.

3.3 ETL/ELT Processing

Development and execution of data transformation processes for cleansing, normalization, and enrichment of Bronze layer data.

- **UTF-8 Encoding** to prevent character encoding issues in team names.
- **Cleansing and standardization** of strings using upper() and trim() on team fields.
- **Type casting:** home_goals and away_goals to IntegerType(), date to TimestampType().
- **Quality filtering:** exclusion of records with goals outside the valid range (0–20).
- **Optimized Parquet writing** with partitioning by the year column.

3.4 Analytical Layer

Configuration of Glue (crawler-silver-parquet) to dynamically map each partition in the Silver layer and expose them in the Data Catalog. SQL queries are executed in Athena to generate the following data per team:

1. Total matches played
2. Wins, draws, and losses
3. Win rate percentage
4. Goals scored and goals conceded

4. Deliverables

The following items are expected to be delivered for validation by Caleb S.A. de C.V.:

Deliverable	Description
Architecture Document	Detailed diagram of AWS services and data flow. Complete technical documentation of the pipeline, AWS services inventory (S3, Glue, Athena, etc.), inter-layer data flow, Gold Layer data dictionary, and execution guide.
Infrastructure Scripts	S3 bucket configuration (unam-2026-ingenieriadatos-equipo6), definition of Glue Job silver1, Crawler crawler-silver-parquet, and associated IAM roles.
Transformation Code	Python/PySpark silver1.py scripts used for cleansing processes (UTF-8, upper, trim), type casting (Integer, Timestamp) with cleansing routines, quality filtering (0-20 goals range), and partitioned Parquet writing.

Deliverable	Description
Results (Gold Layer)	Analytical queries executed in Athena. Interactive dashboard for final data visualization and comparison.

5. Estimated Schedule

Period	Activities
Week 1	Requirements gathering and architecture design.
Week 2	Development of the ingestion pipeline from API-Football to the Bronze Layer. Configuration of Glue Job silver1 and development of the PySpark script silver1.py .
Week 3	Configuration of the Glue Crawler (crawler-silver-parquet), integration with the Data Catalog, and enabling queries in Athena. Generation of results to the Gold Layer.
Week 4	Data quality testing (QA), business rule validation (goals range, encoding, typing, etc.). Cluster performance measurement and final delivery of the corresponding documentation.

6. Acceptance Criteria

The project will be considered complete and accepted when the data in the final layer meets the following business rules and criteria:

1. The data in the Gold Layer matches the business rules defined in the Data Dictionary (correct types, goals range 0–20, fields in uppercase).
2. The Glue Job *silver1* completes the Bronze to Silver transformation in under 2 minutes (reference time measured: 1 min 54 sec on the Serverless cluster).
3. Amazon Athena returns KPI results (wins, draws, losses, win rate, goals) without schema errors.
4. The Glue Crawler correctly updates the Data Catalog partitions after each job run.
5. The Parquet files in the Silver Layer are correctly partitioned by team (/partition_0/) and year (/year=YYYY/)
6. The Gold results are correctly stored at <s3://unam-2026-ingenieriadedatos-equipos6/3gold/>.

7. Signatures and Acceptance

Client	Consulting Team
Accepted by: Caleb S.A. de C.V. Signature: _____ Date: _____	Team 6 — Data Engineering UNAM 2026 Date: _____